

TraceLink

Use Case Descriptions — Updated UCD

Based on the 10 major use cases from Diagram 1 (Updated Overall Use Case Diagram), decomposed using the low-level module diagrams (2–9).

1. Manage Own Account

Use Case ID	UC-01
Use Case Name	Manage Own Account
Primary Actor	SEO Specialist, System Administrator
Brief Description	Allows any authenticated user — SEO Specialist or System Administrator — to manage their own individual account: registering, logging in and out, viewing and updating their profile, changing their password, and archiving their own account.
Preconditions	• For Register Account and Log In: none (these are entry points). • For View Account, Update Account, Change Password, Log Out, and Deactivate or Archive Own Account: the user must already be logged in.
Postconditions – Success	• The account is created, updated, or the session is started/ended as requested.
Postconditions – Failure	• Registration or login is rejected (e.g., invalid credentials or duplicate account).
Main Flow	1. The user registers a new account, or logs in with existing credentials. 2. The system authenticates the user and starts a session. 3. The user views their account details. 4. The user updates account information, if needed. 5. The user changes their password, if needed. 6. The user logs out, ending the session. 7. The user deactivates or archives their own account, if needed.
Alternate / Extension Flows («extend»)	• Reset Forgotten Password «extend» Log In — used when the user cannot remember their password.
Included Use Cases («include»)	Register Account; Log In; View Account; Update Account; Change Password; Log Out; Deactivate or Archive Own Account
Exceptions	• Invalid credentials are submitted — login is rejected. • The registration email or username already exists — registration is rejected.
Notes	• Both the SEO Specialist and System Administrator use this same flow for their own account. • Administering other users' accounts is handled separately under Manage User Accounts.

2. Manage User Accounts

Use Case ID	UC-02
Use Case Name	Manage User Accounts
Primary Actor	System Administrator
Brief Description	Allows the System Administrator to create and manage accounts belonging to other users, including assigning roles, activating or deactivating access, archiving accounts, and resetting user access.
Preconditions	• The System Administrator is logged in.
Postconditions – Success	• The target user account is created, updated, activated/deactivated, archived, or its access is reset, as requested.
Postconditions – Failure	• The requested change is rejected (e.g., duplicate account or an invalid role assignment).
Main Flow	8. The System Administrator creates a new user account. 9. The System Administrator views a user account's details. 10. The System Administrator updates a user account. 11. The System Administrator assigns a user role. 12. The System Administrator activates or deactivates a user account. 13. The System Administrator archives a user account, if needed. 14. The System Administrator resets a user's access, if requested.
Alternate / Extension Flows («extend»)	None defined in the source diagram.
Included Use Cases («include»)	Create User Account; View User Account; Update User Account; Assign User Role; Activate or Deactivate User Account; Archive User Account; Reset User Access
Exceptions	• The account or email already exists — creation is rejected. • An invalid role is assigned — the assignment is rejected.
Notes	• The SEO Specialist is not connected to this use case — managing other users' accounts is exclusive to the System Administrator.

3. Manage URL Batches

Use Case ID	UC-03
Use Case Name	Manage URL Batches
Primary Actor	SEO Specialist
Brief Description	Allows the SEO Specialist to upload a dataset of URLs, clean and validate the data, and confirm a batch so it is ready for scanning.
Preconditions	• The SEO Specialist is logged in.
Postconditions – Success	• A validated, de-duplicated batch of URLs is confirmed and available for Run Batch Scan.
Postconditions – Failure	• The file upload is rejected, or the batch is left unconfirmed and remains in preview.
Main Flow	15. The SEO Specialist uploads a URL file (CSV or Excel). 16. The system validates the file type and size. 17. The system extracts the URL records from the file. 18. The system selects the appropriate URL column. 19. The system normalizes the extracted URLs. 20. The system detects invalid URLs. 21. The system detects duplicate URLs. 22. The system displays a preview of the batch. 23. The SEO Specialist reviews the preview and confirms the batch. 24. The system saves the confirmed batch and makes it available for scanning.
Alternate / Extension Flows («extend»)	• Select URL Column «extend» Upload URL File — used when the file has multiple possible URL columns. • Resolve Invalid Rows «extend» Preview Batch — used when invalid rows are detected and the specialist fixes or excludes them. • Remove Duplicate Entries «extend» Preview Batch — used when duplicates are found and the specialist chooses to remove them.
Included Use Cases («include»)	Upload URL File; Validate File Type and Size; Extract URL Records; Select URL Column; Normalize URLs; Detect Invalid URLs; Detect Duplicate URLs; Preview Batch; Confirm Batch; View Batch Details; Delete or Archive Batch
Exceptions	• The uploaded file has an unsupported type or exceeds the size limit — upload is rejected. • No valid URLs remain after validation — the batch cannot be confirmed.
Notes	• The System Administrator is connected to this use case only if administrators are formally allowed to upload and scan files.

4. Run Batch Scan

Use Case ID	UC-04
Use Case Name	Run Batch Scan
Primary Actor	SEO Specialist
Secondary Actor(s)	External Website, AI Model Service
Brief Description	Executes the full scan of a confirmed URL batch: checking cached results, validating URL format, contacting each site, following redirects, retrieving accessible page content, analyzing webpage structure and content (including AI-assisted relevance/quality evaluation), and saving results for review, monitoring, and reporting.
Preconditions	• The SEO Specialist is logged in. • A URL batch has already been confirmed through Manage URL Batches.
Postconditions – Success	• Every URL in the batch has a saved technical scan result and, where applicable, an AI analysis result. • URLs the system could not confidently classify are queued for manual review. • Scan progress is marked complete.
Postconditions – Failure	• The scan job is cancelled or interrupted; partially processed URLs retain a saved result and the remainder stay queued or marked as failed.
Main Flow	25. The SEO Specialist selects a confirmed URL batch and starts the scan. 26. The system creates a scan job for the batch. 27. The system queues the URL records for processing. 28. For each URL, the system checks whether a cached result exists. 29. The system validates the URL format. 30. The system sends an HTTP request to the URL (External Website). 31. The system classifies the HTTP response. 32. If the response is a redirect, the system follows the redirect chain and records the final destination. 33. The system retrieves the accessible webpage content (External Website). 34. The system analyzes the accessible webpage — extracting the page title, metadata, headings, links, forms, input fields, and comment sections, then extracting the main content and preparing it for AI analysis. 35. The system submits the prepared content to the AI Model Service for relevance and quality evaluation, and records the AI analysis result. 36. The system saves the technical scan result. 37. The system updates the scan progress. 38. Steps 4–13 repeat until every URL in the batch has been processed. 39. The system marks the scan job complete and notifies the SEO Specialist.
Alternate / Extension Flows («extend»)	• Reuse Valid Cached Result «extend» Check Cached Result — used when a fresh cached result already exists for the URL. • Refresh Expired Cached Result «extend» Check Cached Result — used when the cached result has passed its freshness period. • Follow Redirect Chain «extend» Classify HTTP Response — used when the HTTP response is a redirect status.

	● Retry Rate-Limited URL «extend» Handle URL Error — used when the target site returns a rate-limit response. ● Mark for Manual Review «extend» Handle URL Error — used when the URL cannot be classified automatically (e.g., CAPTCHA, login wall). ● Cancel Scan «extend» Monitor Scan — used when the SEO Specialist cancels an in-progress scan. ● Retry Failed URLs «extend» View Failed Scan Results — used when the SEO Specialist retries URLs after reviewing failures. ● Truncate or Chunk Content «extend» Prepare Content for AI Analysis — used when extracted content exceeds the AI model's input limit. ● Mark Partial Extraction «extend» Extract Main Page Content — used when content extraction is incomplete. ● Mark AI Evaluation Failure «extend» Evaluate Content Relevance — used when the AI Model Service cannot return a result. ● Queue for Manual Review «extend» Record AI Analysis Result — used when the AI result is ambiguous or low-confidence.
Included Use Cases («include»)	Create Scan Job; Queue URL Records; Check Cached Result; Validate URL Format; Send HTTP Request; Classify HTTP Response; Follow Redirect Chain; Record Final Destination; Handle URL Error; Retrieve Accessible Webpage; Analyze Accessible Webpage (Extract Page Title, Extract Metadata, Extract Headings, Detect Internal and External Links, Detect Forms, Detect Input Fields, Detect Comment Sections, Extract Main Page Content, Prepare Content for AI Analysis, Evaluate Content Relevance, Evaluate Content Quality, Detect Spam-Like Indicators, Record AI Analysis Result); Save Technical Scan Result; Update Scan Progress
Exceptions	● The URL is unreachable or times out — status is recorded as "unreachable" and Handle URL Error is triggered. ● The target site returns a rate-limit response — Retry Rate-Limited URL extension is triggered. ● Content cannot be classified automatically — Mark for Manual Review extension is triggered. ● The AI Model Service fails to return a classification — Mark AI Evaluation Failure extension is triggered.
Notes	● External Website is associated only with Send HTTP Request, Follow Redirect Chain, and Retrieve Accessible Webpage. ● AI Model Service is associated only with Evaluate Content Relevance, Evaluate Content Quality, and Detect Spam-Like Indicators. ● The SEO Specialist is connected only to starting/monitoring the scan — not directly to internal HTTP-validation or content-analysis steps.

5. Monitor Scan

Use Case ID	UC-05
Use Case Name	Monitor Scan
Primary Actor	SEO Specialist
Brief Description	Allows the SEO Specialist to track the real-time progress of an active batch scan and intervene — cancelling the job or retrying failed URLs — while it runs.
Preconditions	• The SEO Specialist is logged in. • A batch scan (Run Batch Scan) is currently in progress.
Postconditions – Success	• The specialist has current visibility into scan progress, and any cancel or retry action taken is applied to the job.
Postconditions – Failure	• A requested cancel or retry action fails to apply (e.g., the job has already completed).
Main Flow	40. The SEO Specialist opens the monitoring view for an active scan job. 41. The system displays scan progress, including queued, in-progress, completed, cached, and failed URL counts. 42. The system continuously updates the progress display as the scan runs. 43. The SEO Specialist reviews failed scan results, if any. 44. The SEO Specialist cancels the scan or retries failed URLs, if needed. 45. The system applies the requested action and updates the job status accordingly.
Alternate / Extension Flows («extend»)	• Cancel Scan «extend» Monitor Scan — used when the specialist stops an in-progress scan job. • Retry Failed URLs «extend» View Failed Scan Results — used when the specialist re-queues URLs that previously failed.
Included Use Cases («include»)	View Failed Scan Results
Exceptions	• A cancel or retry request is submitted for a job that has already completed — the system shows an error and takes no action.
Notes	• The source document does not define a full «include» breakdown for this use case the way it does for the other modules — this description is reconstructed from the asynchronous-processing requirements described elsewhere in the document (progress visibility, pending/failed/cached counts, cancel, retry) plus the two «extend» relationships defined under Run Batch Scan. Recommend the team formalize a dedicated diagram for this use case before finalizing.

6. Perform Manual Review

Use Case ID	UC-06
Use Case Name	Perform Manual Review
Primary Actor	SEO Specialist
Brief Description	Allows the SEO Specialist to review URLs the system could not confidently classify — such as CAPTCHA-protected, blocked, or ambiguous results — and record a final decision.
Preconditions	• The SEO Specialist is logged in. • One or more URLs have been flagged for manual review during Run Batch Scan.
Postconditions – Success	• The reviewed URL has a saved manual review decision (approved, corrected, or rejected), with optional notes.
Postconditions – Failure	• The URL remains in the review queue, unresolved.
Main Flow	46. The SEO Specialist opens the review queue. 47. The system displays the URLs pending manual review. 48. The SEO Specialist opens a URL's review details. 49. The system displays the technical validation result, extracted page elements, and AI classification for that URL. 50. The SEO Specialist evaluates the classification. 51. The SEO Specialist approves, corrects, or rejects the result. 52. The SEO Specialist optionally adds review notes. 53. The system saves the manual review decision and removes the URL from the queue.
Alternate / Extension Flows («extend»)	• Open Original Webpage «extend» Open URL Review Details — used when the specialist wants to view the live page. • Correct Relevance Label «extend» Correct Classification — used when only the relevance label is wrong. • Correct Quality Label «extend» Correct Classification — used when only the quality label is wrong. • Rescan URL «extend» Open URL Review Details — used when the specialist requests a fresh scan instead of deciding manually.
Included Use Cases («include»)	View Review Queue; Open URL Review Details; View Technical Validation Result; View Extracted Page Elements; View AI Classification; Approve Classification; Correct Classification; Reject Result; Add Review Notes; Save Manual Review Decision
Exceptions	• A requested rescan fails or times out — the URL remains flagged and the error is logged.
Notes	• The System Administrator does not perform SEO judgment on flagged content unless the project explicitly assigns that responsibility.

7. View Scan Results

Use Case ID	UC-07
Use Case Name	View Scan Results
Primary Actor	SEO Specialist
Secondary Actor(s)	System Administrator
Brief Description	Allows an actor to view processed scan results at a summary and URL level, including technical, structural, AI, and cache details, with filtering, sorting, and search.
Preconditions	At least one batch scan has completed and produced results.
Postconditions – Success	The requested results are displayed to the actor.
Postconditions – Failure	No results match the applied filter, sort, or search criteria.
Main Flow	54. The actor opens the scan results view. 55. The system displays the scan summary. 56. The system displays URL-level results, including HTTP status, redirect information, page structure findings, AI analysis, cache status, and error details. 57. The actor filters, sorts, or searches the results as needed.
Alternate / Extension Flows («extend»)	View Full URL Details «extend» View URL-Level Results — used when the actor opens an expanded view of a single URL.
Included Use Cases («include»)	View Scan Summary; View URL-Level Results; View HTTP Status; View Redirect Information; View Page Structure Findings; View AI Analysis; View Cache Status; View Error Details; Filter Results; Sort Results; Search Results
Exceptions	No data matches the applied filter — an empty state is shown.
Notes	This use case is a prerequisite for Generate and Export Reports — a report is compiled from the results viewed here.

8. Generate and Export Reports

Use Case ID	UC-08
Use Case Name	Generate and Export Reports
Primary Actor	SEO Specialist
Secondary Actor(s)	System Administrator
Brief Description	Allows an actor to compile scan results into a report and export that report in a chosen file format.
Preconditions	• Scan results are available to report on (see View Scan Results).
Postconditions – Success	• A report is generated and successfully exported/downloaded in the selected format.
Postconditions – Failure	• Report generation or export fails, or the actor cancels before completing the export.
Main Flow	58. The actor selects a report scope. 59. The system generates summary metrics and the URL-level report, including manual review decisions and processing metrics. 60. The actor selects an export format (CSV or Excel). 61. The system exports the report. 62. The actor downloads the report file.
Alternate / Extension Flows («extend»)	• Export Filtered Results «extend» Export Report — used when the actor exports only the currently filtered subset of results.
Included Use Cases («include»)	Select Report Scope; Generate Summary Metrics; Generate URL-Level Report; Include Manual Review Decisions; Include Processing Metrics; Select Export Format; Export as CSV; Export as Excel; Download Report
Exceptions	• The requested export format is unsupported, or file generation fails — an error is shown and the actor may retry.
Notes	• The source document later splits this into two separate parent use cases (Generate Report and Export Report) at the low-level diagram stage. They are kept combined here to match the 10-item major use case list; the team should decide which grouping is final. • A Business Representative actor should only be added if the team formally decides that report-only users will have system accounts.

9. Manage Cache and Integrity

Use Case ID	UC-09
Use Case Name	Manage Cache and Integrity
Primary Actor	System Administrator
Brief Description	Allows the System Administrator to monitor and maintain the cache used by the scanning process — reviewing statistics, configuring freshness rules, and resolving data-integrity issues.
Preconditions	• The System Administrator is logged in. • Cached scan data exists in the system.
Postconditions – Success	• Cache configuration is updated, stale or invalid records are purged, or a corrupted record is successfully reprocessed.
Postconditions – Failure	• The requested change is rejected (e.g., an invalid configuration value), or a record fails to be reprocessed.
Main Flow	63. The System Administrator views cache statistics. 64. The System Administrator configures the cache freshness period, if a change is needed. 65. The System Administrator searches cache records. 66. The System Administrator purges selected cache records. 67. The System Administrator reviews integrity alerts. 68. The System Administrator verifies stored record integrity. 69. The System Administrator views integrity logs. 70. The System Administrator reprocesses a corrupted record, if needed.
Alternate / Extension Flows («extend»)	• Reprocess Corrupted Record «extend» Review Integrity Alerts — used when an alert indicates a corrupted record. • Purge Expired Records «extend» Search Cache Records — used when the administrator bulk-removes expired records found in a search.
Included Use Cases («include»)	View Cache Statistics; Configure Cache Freshness Period; Search Cache Records; Purge Cache Records; Review Integrity Alerts; Verify Stored Record Integrity; View Integrity Logs; Reprocess Corrupted Record
Exceptions	• Reprocessing a corrupted record fails — the record stays flagged as corrupted and the error is logged.
Notes	• Automatic, scan-triggered cache behavior (Process Cached Scan Data: generating fingerprints, checking freshness, reusing valid results) runs as part of Run Batch Scan and has no actor — it is not part of this administrator-facing use case.

10. Monitor System Operations

Use Case ID	UC-10
Use Case Name	Monitor System Operations
Primary Actor	System Administrator
Brief Description	Allows the System Administrator to observe overall system health and job activity — active, completed, and failed scan jobs, error logs, API usage, processing metrics, storage usage, and user activity — and intervene on stuck or failed jobs.
Preconditions	• The System Administrator is logged in.
Postconditions – Success	• The administrator has current visibility into system status, and any retry or cancel action taken is applied to the affected job.
Postconditions – Failure	• A requested retry or cancel action fails to apply.
Main Flow	71. The System Administrator views active scan jobs. 72. The System Administrator views completed scan jobs. 73. The System Administrator views failed scan jobs. 74. The System Administrator views error logs. 75. The System Administrator views API usage. 76. The System Administrator views processing metrics. 77. The System Administrator views storage usage. 78. The System Administrator views user activity logs. 79. The System Administrator retries a failed job or cancels a stuck job, if needed.
Alternate / Extension Flows («extend»)	• Retry Failed Job «extend» View Failed Scan Jobs — used when the administrator re-queues a failed job. • Cancel Stuck Job «extend» View Active Scan Jobs — used when the administrator terminates a job that is no longer progressing. • Review API Limit Warning «extend» View API Usage — used when API usage approaches a configured limit.
Included Use Cases («include»)	View Active Scan Jobs; View Completed Scan Jobs; View Failed Scan Jobs; View Error Logs; View API Usage; View Processing Metrics; View Storage Usage; View User Activity Logs; Retry Failed Job; Cancel Stuck Job
Exceptions	• A retry or cancel request is submitted for a job in an invalid state — the system shows an error and takes no action.
Notes	• This use case is exclusive to the System Administrator; the SEO Specialist has no visibility into system-wide operations.